EXAMPLE 15

Wall Object Behavior - Static Lateral Loads Analysis

Problem Description

This example analyzes a series of wall configurations to evaluate the behavior of the ETABS shell object with wall section assignments. All walls are subjected to a static lateral load applied at the top of the wall.

The following walls are included:

- Planar shear wall, shown in Figure 15-1
- Wall supported on columns, shown in Figure 15-2
- Wall-spandrel system, shown in Figure 15-3
- C-shaped wall section, shown in Figure 15-4
- Wall with edges thickened, shown in Figure 15-5
- E-shaped wall section, shown in Figure 15-6

Geometry, Properties and Loads

A modulus of elasticity of 3000 ksi and a Poisson's ratio of 0.2 are used for all walls. Kip-inch-second units are used throughout. The following sections describe the models for the different walls.

Planar Shear Wall , Example 15a
This shear wall is modeled with one panel per story. Three different wall lengths of 120", 360" and 720" are analyzed. Also, one-story and three-story walls are analyzed, together with the six-story wall shown in Figure 15-1. A wall thickness of 12" is used.

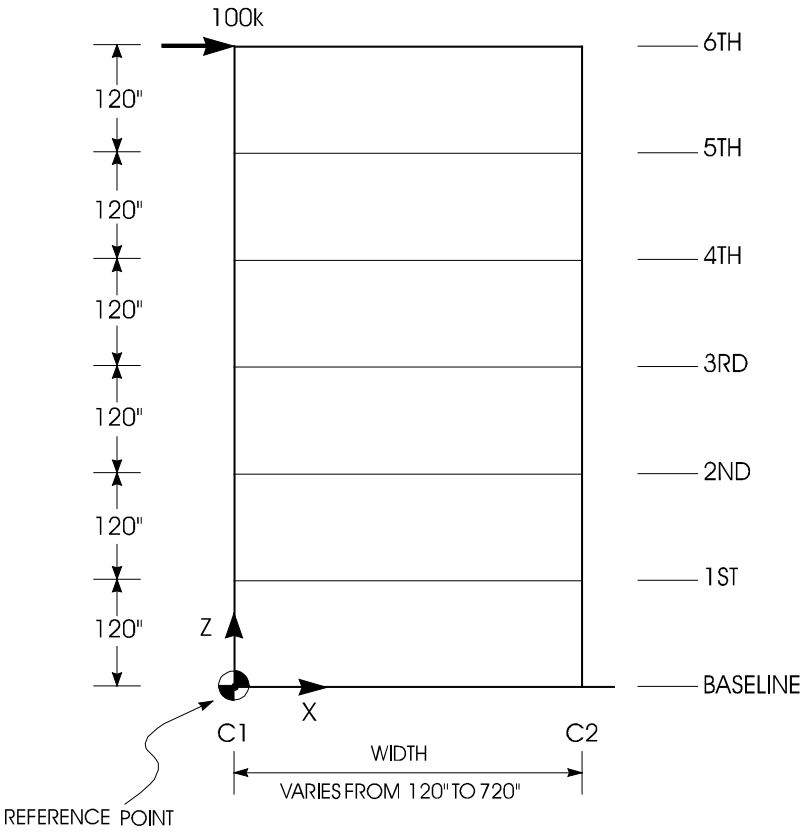


Figure 15-1 Planar Shear Wall, Example 15a

Wall Supported on Columns, Example 15b

This wall is modeled with two column lines. Columns are used for the first story, and the top two stories have a single shell object with end piers, as shown in Figure 15-2. End piers are 40" by 12" in cross section and panels are 12" thick. Columns are 40" by 20" in cross section.

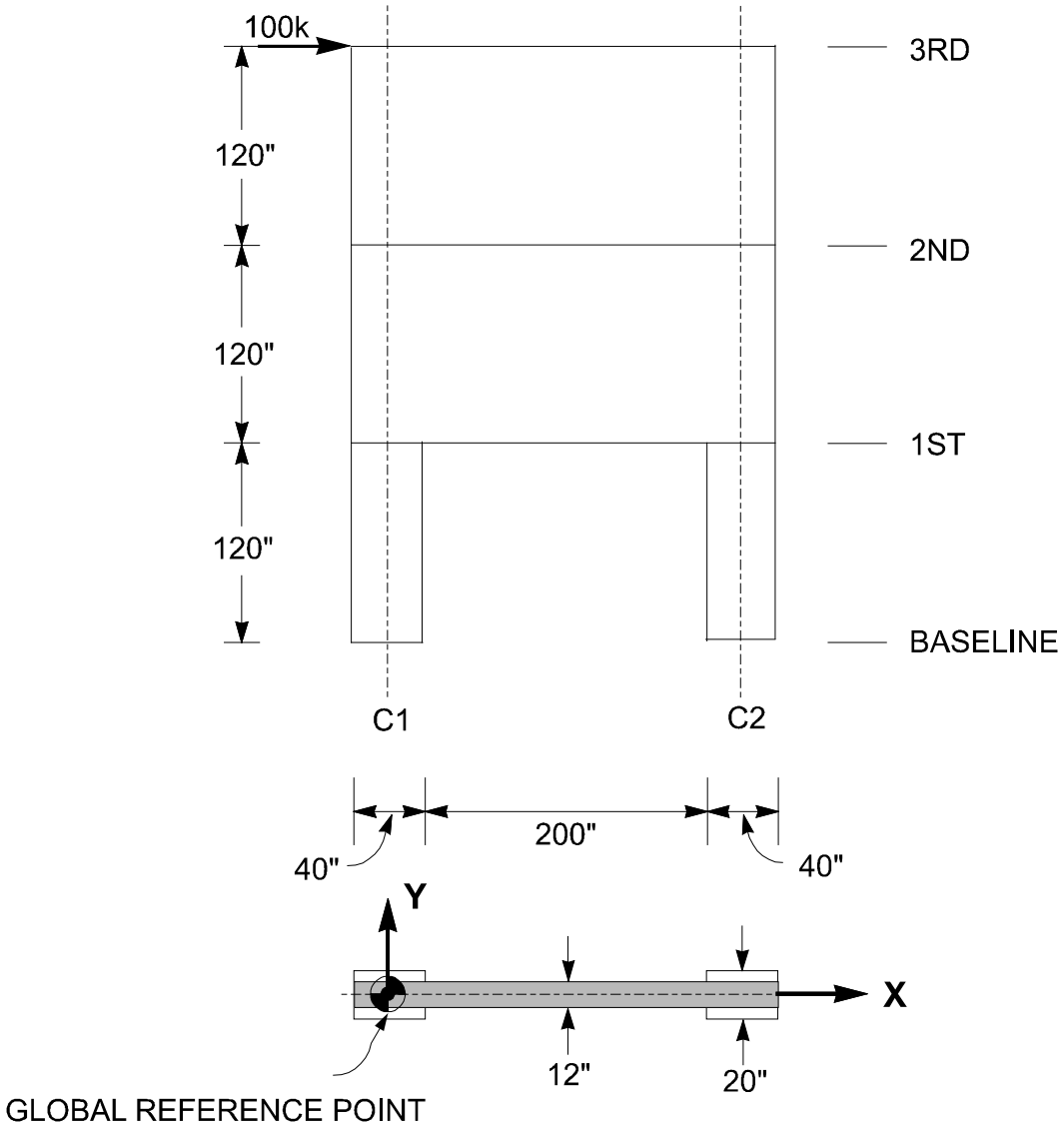


Figure 15-2 Wall Supported on Columns, Example 15b

Wall-Spandrel System, Example 15c

This wall is modeled with four column lines. The spandrels are modeled as beams. Two different spandrel lengths of 60" and 240" are analyzed. Each wall is modeled with two shell objects per story. Three-story walls are also analyzed together with the six-story wall shown in Figure 15-3. A wall and spandrel thickness of 12" is used.

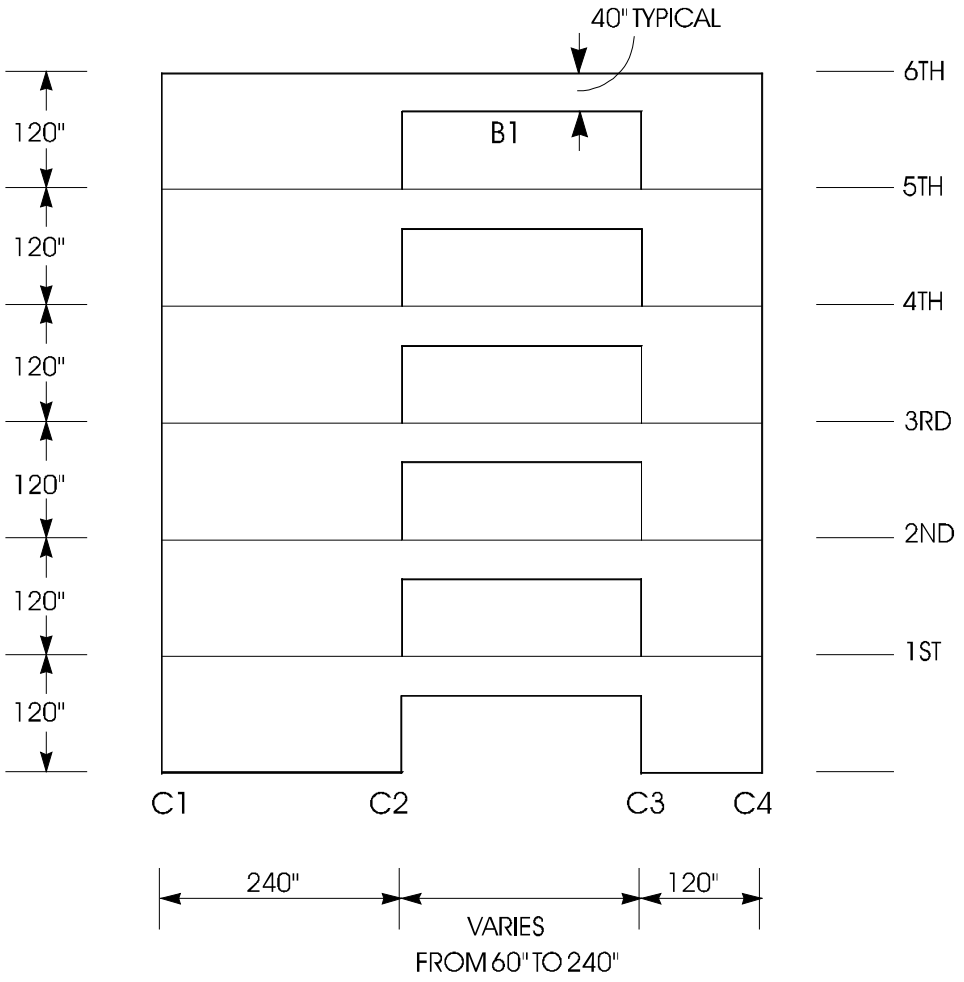


Figure 15-3 Wall-Spandrel System, Example 15c

Shaped Wall Section, Example 15d

This wall is modeled with six column lines and five shell objects per story, to model the shape of the wall. A three-story wall was also analyzed together with the six-story wall, as shown in Figure 15-4. A wall thickness of 6" is used.

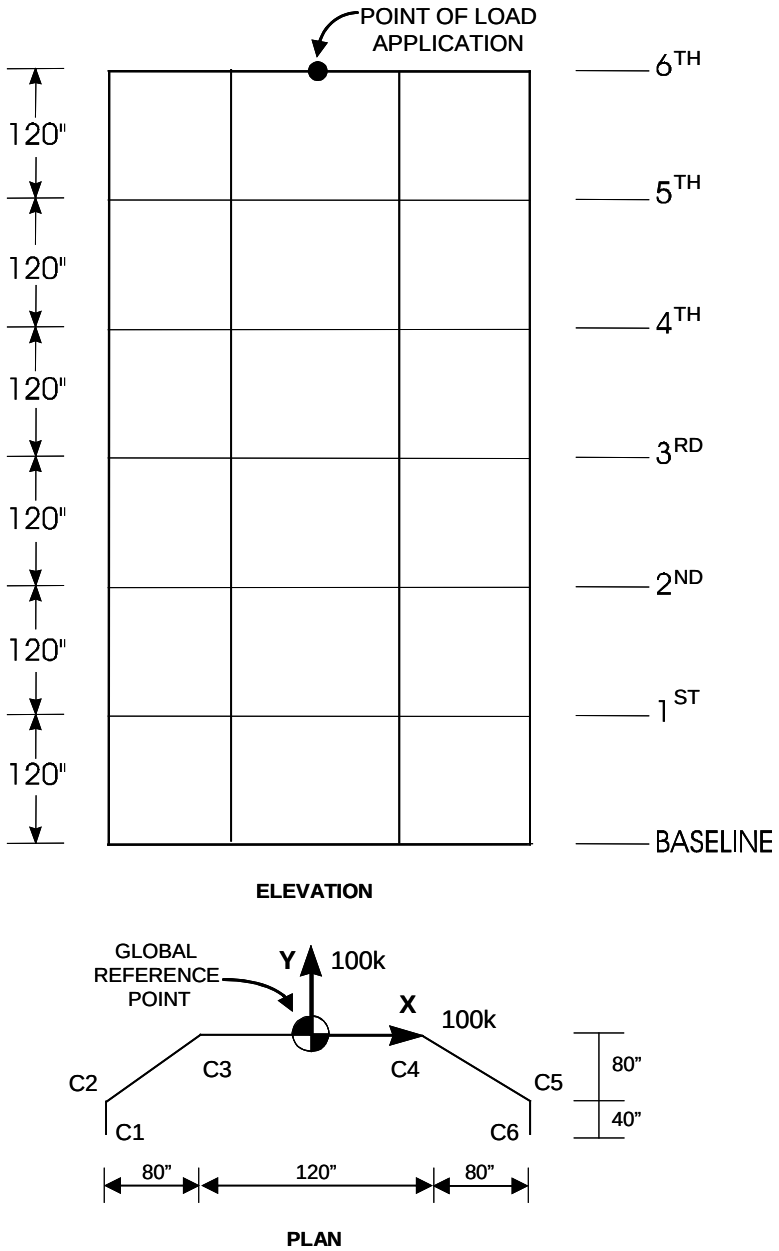


Figure 15-4 C-Shaped Wall Section, Example 15d

Wall with Edges Thickened, Example 15e

This wall is modeled with two column lines and one shell object, with end piers, per story as shown in Figure 15-5. A three-story wall was also analyzed together with the six-story wall shown in Figure 15-5.

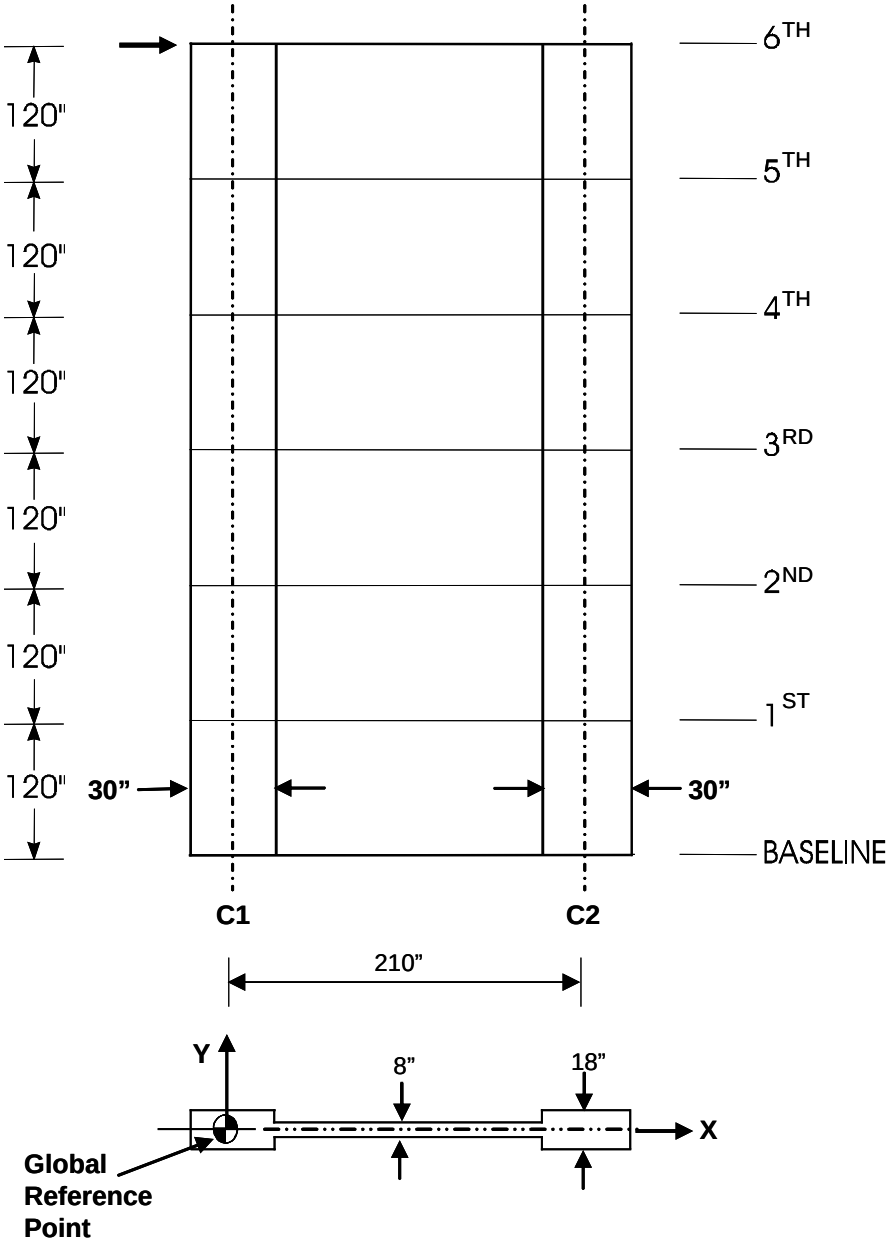


Figure 15-5 Wall with Thickened Edges, Example 15e

E-Shaped Wall Section, Example 15f

This wall is modeled with six column lines and five shell objects per story to model the shape of the wall. A three-story wall was also analyzed together with the six-story wall, as shown in Figure 15-6. A wall thickness of 6" is used.

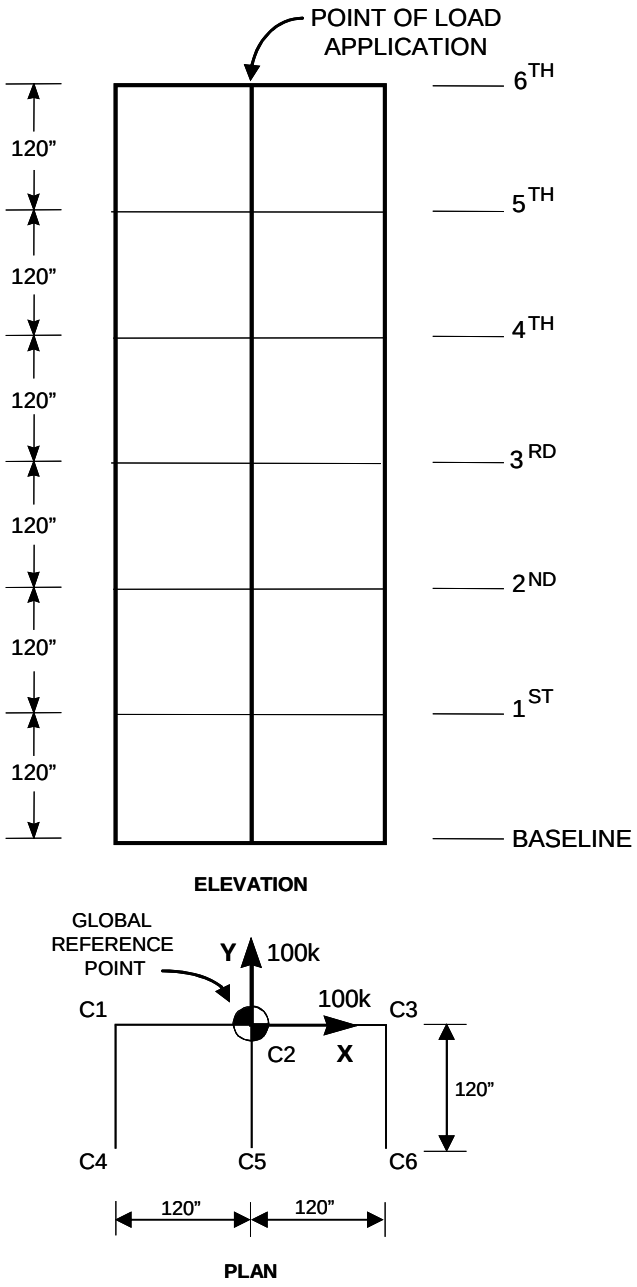


Figure 15-6 E-Shaped Wall Section, Example 15f

Technical Features of ETABS Tested

- Use of area objects
- Two-dimensional and three-dimensional shear wall systems
- Static lateral loads analysis

Results Comparison

All walls analyzed in this example using ETABS were also analyzed using the general structural analysis program SAP2000 (Computers and Structure 2002), using refined meshes of the membrane/shell element of that program. The SAP2000 meshes used are shown in Figures 15-7, 15-8, 15-9, 15-10, 15-11 and 15-12. For the SAP2000 analysis, the rigid diaphragms at the floor levels were modeled by constraining all wall nodes at the floor to have the same lateral displacement for planar walls, or by adding rigid members in the plane of the floor for three-dimensional walls.

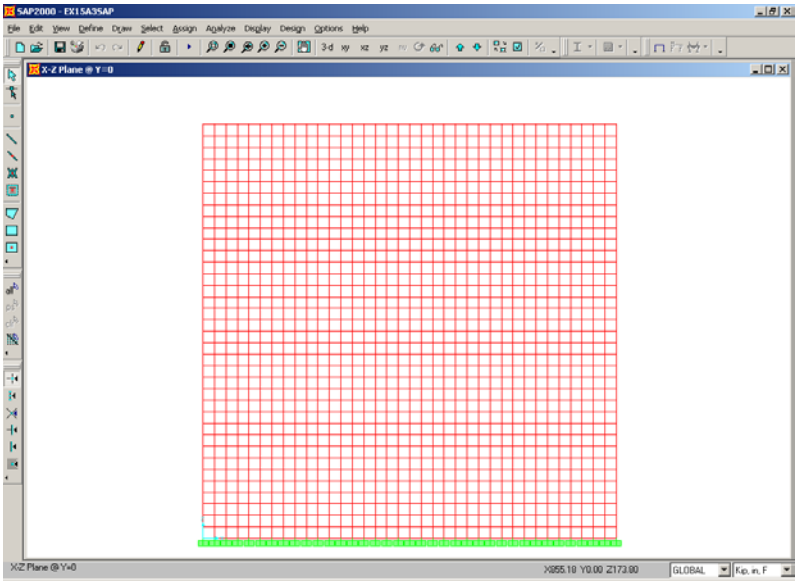


Figure 15-7 SAP2000 Mesh, Example 15a

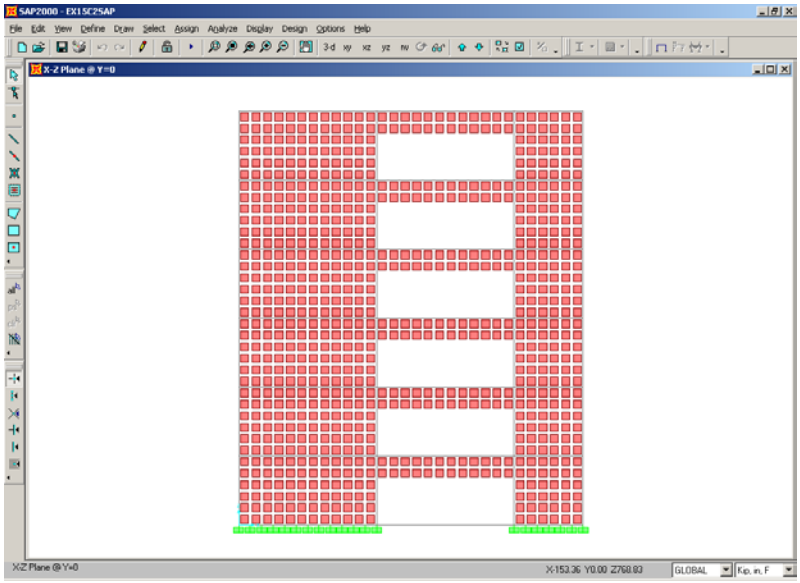


Figure 15-8 SAP2000 Mesh, Example 15b

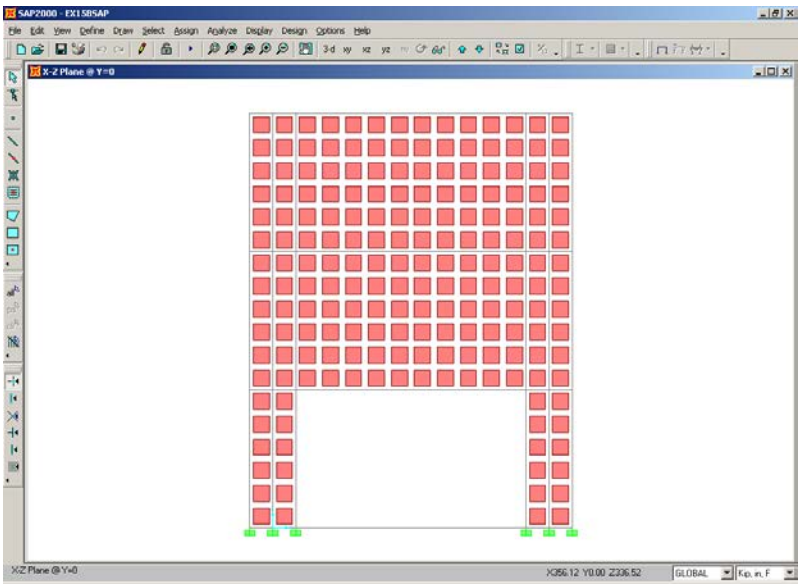


Figure 15-9 SAP2000 Mesh, Example 15c

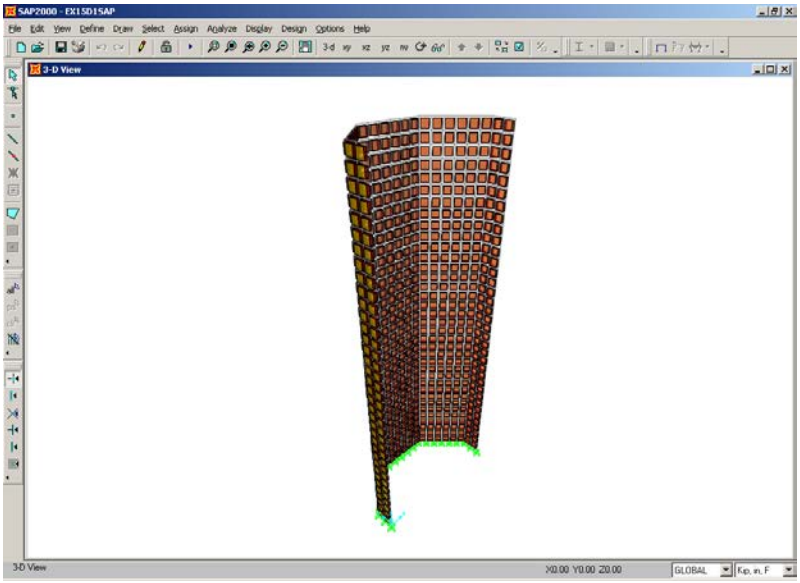


Figure 15-10 SAP2000 Mesh, Example 15d

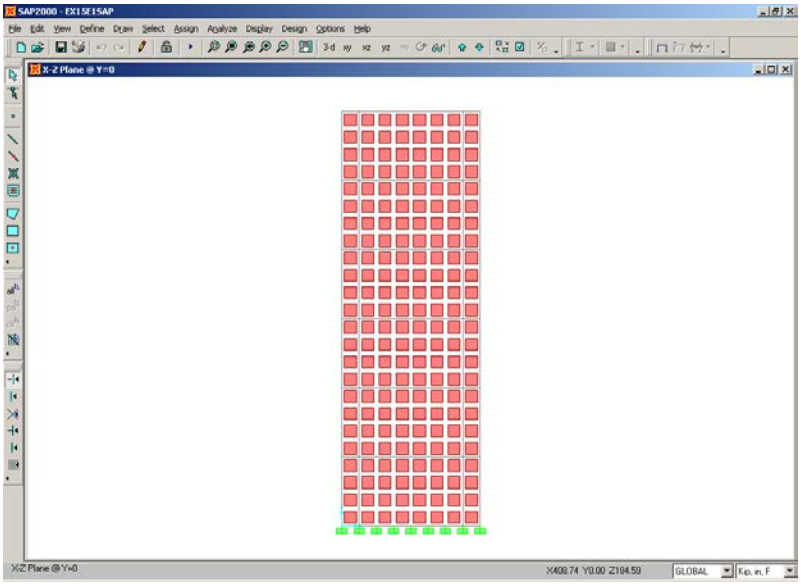


Figure 15-11 SAP2000 Mesh, Example 15e

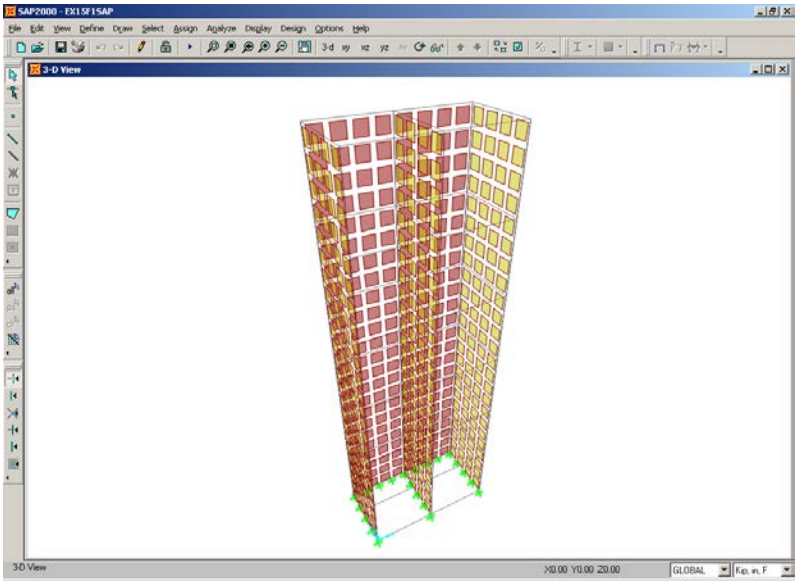


Figure 15-12 SAP2000 Mesh, Example 15f

The lateral displacements from the ETABS and SAP2000 analyses are compared in Tables 15-1, 15-2, 15-3, 15-4, 15-5 and 15-6 for the various walls.

Table 15-1 Results Comparison for Top Displacements (Inches), Example 15a

Number of Stories	Wall Height (inches)	Wall Length (inches)	ETABS	SAP2000
6	720	120	2.3921	2.4287
		360	0.0986	0.1031
		720	0.0172	0.0186
3	360	120	0.3071	0.3205
		360	0.0170	0.0187
		720	0.0046	0.0052
1	120	120	0.0145	0.0185
		360	0.0025	0.0029
		720	0.0011	0.0013

Table 15-2 Results Comparison for Displacements (Inches), Example 15b

Location	ETABS	SAP2000
Story 3	0.0691	0.0671
Story 2	0.0524	0.0530
Story 1	0.0390	0.0412

Table 15-3 Results Comparison for Top Displacements (inches) Example 15c (1-4)

Number of Stories	Beam Length (inches)	ETABS	SAP2000
6	60	0.0844	0.0869
	240	0.1456	0.1505
3	60	0.0188	0.0200
	240	0.0313	0.0332

Table 15-4 Results Comparison for Top Displacements (Inches) at Load Application Point, Example 15d (1-2)

Number of Stories	Load Direction	Displacement Direction	ETABS	SAP2000
6	X	X	0.8637	0.8936
	X	Z-Rotation	0.0185	0.0191
	Y	Y	1.1447	1.1882
3	X	X	0.1249	0.1337
	X	Z-Rotation	0.0024	0.0025
	Y	Y	0.1623	0.1733

Table 15-5 Results Comparison for Top Displacements (Inches), Example 15e(1-2)

Number of Stories	ETABS	SAP2000
6	0.2822	0.2899
3	0.0464	0.0480

Table 15-6 Results Comparison for Displacements at Load Application, Example 15f (1-2)

Number of Stories	Load Direction	Displacement Direction	ETABS	SAP2000
6	X	X	0.3707	0.3655
	X	Z-Rotation	0.0042	0.0039
	Y	Y	0.7295	0.7490
3	X	X	0.0602	0.0628
	X	Z-Rotation	0.0005	0.0005
	Y	Y	0.0993	0.1058

Computer Files

The input data files for the planar shear walls are included as files Example 15A1.EDB through Example 15A9.EDB. These and the following input data files are provided as part of the ETABS installation.

The input data for the wall supported on columns is Example 15B.EDB.

The input data files for the wall-spandrel system are Example C1.EDB through Example C4.EDB.

The input data files for the shaped wall section are included as files Example 15D1.EDB and Example 15D2.EDB.

The input data for the wall with thickened edges are included as files Example 15E1.EDB and Example 15E2.EDB.

The input data for the E-shaped wall section are included as files Example 15F1.EDB and Example 15F2.EDB.

Conclusion

The results comparison show acceptable agreement between ETABS and SAP2000. In general, the comparisons become better as the number of stories increases.